

prepare_fold_spatial_predictors.R

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prepare_fold_spatial_predictors
Prepare Fold-Local Spatial Predictors

Description

Computes Moran eigenvector predictors from training locations or samples only and projects them to held-out samples with the production IDW path.

Usage

```
prepare_fold_spatial_predictors(  
  data_coords_projected = NULL,  
  data_sample_ids = NULL,  
  train_ids = NULL,  
  test_ids = NULL,  
  spatial_mode = NULL,  
  n_mev = NULL,  
  spatial_mev_config = NULL,  
  compute_spatial_function = compute_spatial_mev,  
  compute_spatiotemporal_function = compute_spatiotemporal_mev,  
  interpolate_spatial_function = interpolate_mev_to_grid,  
  interpolate_spatiotemporal_function = interpolate_st_mev_to_grid,  
  compute_spatial_basis_function = compute_spatial_mev_basis,  
  project_spatial_basis_function = project_spatial_mev_basis  
)
```

Arguments

`data_coords_projected`
Data frame with site row names and 'coord_x_km' and 'coord_y_km' columns.

`data_sample_ids`
Data frame containing 'dataset_name' and 'age' for all candidate samples.

`train_ids, test_ids`
 Disjoint sample identifiers in "dataset_name__age" format. Locations represented by the two partitions must also be disjoint.

`spatial_mode` Character scalar. Supported values are "spatial" and "spatiotemporal".

`n_mev` Positive integer giving the requested number of Moran eigenvectors.

`spatial_mev_config`
 Optional shared 2-D MEM strategy configuration. When 'NULL', the legacy compute and interpolation functions are retained for compatibility tests.

`compute_spatial_function, compute_spatiotemporal_function`
 Functions used to compute 2-D and spatiotemporal Moran eigenvectors.

`interpolate_spatial_function, interpolate_spatiotemporal_function`
 Functions used to project training Moran eigenvectors to held-out samples.

`compute_spatial_basis_function, project_spatial_basis_function`
 Functions used to construct and project an explicit reusable 2-D basis.

Value

Named list with unscaled training and held-out spatial predictor data frames plus fold-level spatial diagnostics, including requested, available, and unavailable Moran eigenvector counts.

Examples

```
## Not run:
list_spatial_fold <-
  prepare_fold_spatial_predictors(
    data_coords_projected = data_coords,
    data_sample_ids = data_samples,
    train_ids = vec_train_ids,
    test_ids = vec_test_ids,
    spatial_mode = "spatial",
    n_mev = 3L
  )

## End(Not run)
```

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